



DEALER BEST PRACTICE SERIES

3500, 3400, C175
Crankshaft Turn Tool for
Short Block Assembly

Application Maintenance Component Rebuild Component Life Management MARC Management

3500, 3400, C175 Crankshaft Turn Tool for Short Block Assembly 0

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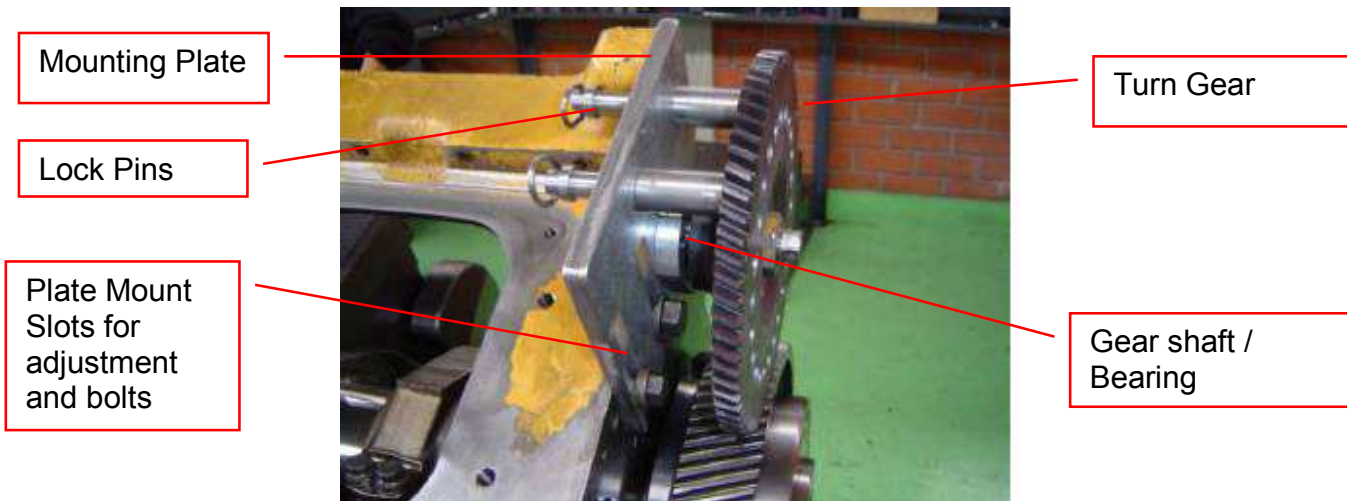
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1.0 Introduction

There is no Caterpillar tool to rotate a 3500 series engine crankshaft in a short block when the engine front and rear covers are no installed. It is necessary to turn the crankshaft in order to install the connecting rods. This Best Practice is a dealer fabricated tool to rotate the crankshaft. The tool also locks into position so the crankshaft does not move while the connecting rod bolts are torqued.

2.0 Best Practice Description

This Best Practice has been implemented by Ferreyros.



This tool makes it easy to rotate the crankshaft to any desired position and then lock the crankshaft in place. This makes the installation of connecting rods easier and assures that the crankshaft will not slip during the connecting rod bolt torque process.



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Camshaft BEARING Installation Tool

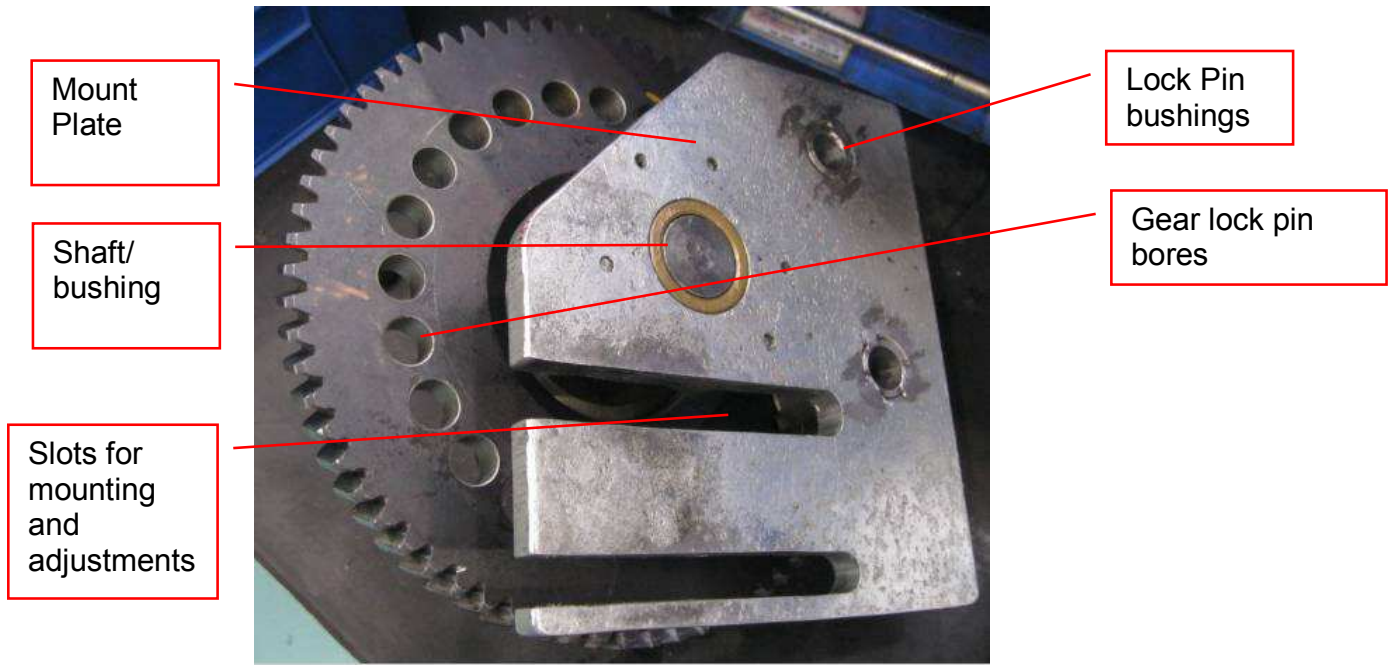
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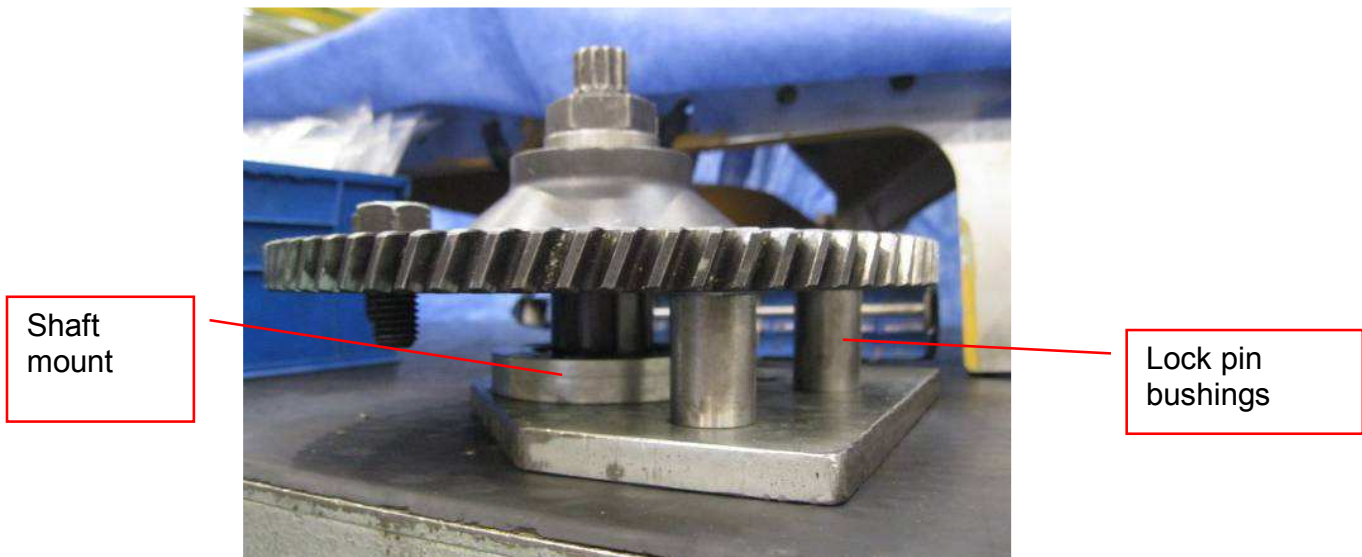
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Since no print drawings are available at this time, the following photos show the tool construction.

Back View



Side View



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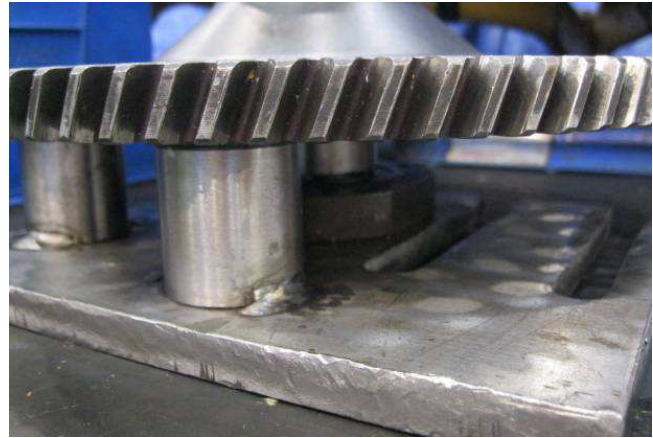
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Top View



Side View



Turning tool for a 3400 series engine is similar in concept. Photo below.



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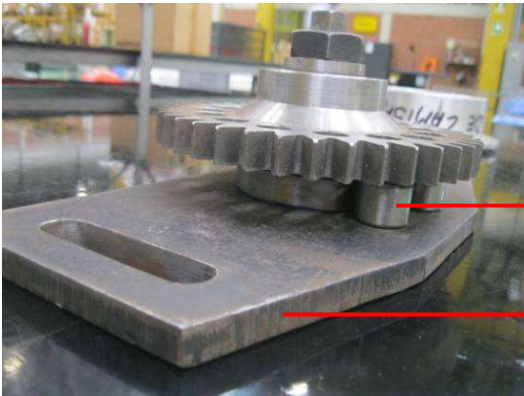
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Turning tool for C175 engine consists of the following parts:



Lock Pin Bushings

Mounting Plate



Turn Gear



Gear shaft / Bearing

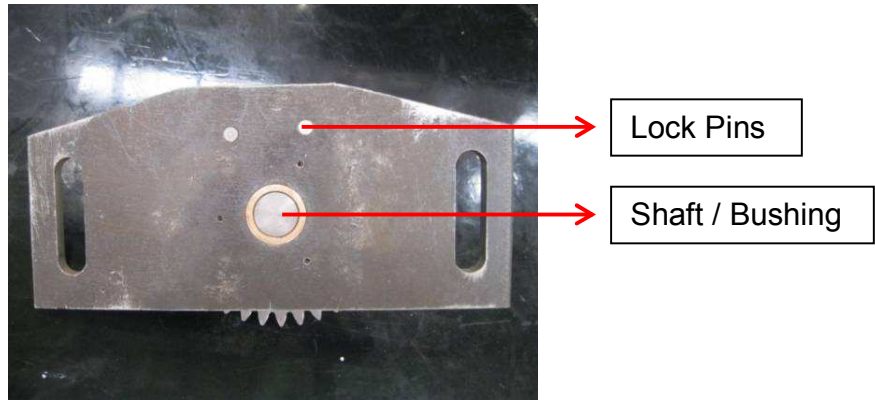
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This tool is installed in the block next to the crankshaft gear and allows a turn clockwise and counterclockwise to the desired location by the mechanic.



Reference Picture

3.0 Implementation Steps

- Design tool with desired features
- Fabricate tool

4.0 Benefits

- Easy for technician to rotate crankshaft.
- Improves the quality of the connecting rod bolt torque process
- Improves short block assembly efficiency

5.0 Resources Required

6.0 Supporting Attachments / References

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7.0 Related Best Practices

8.0 Acknowledgements

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May 29, 2012 – Document updated to include tool for C175 engine.

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